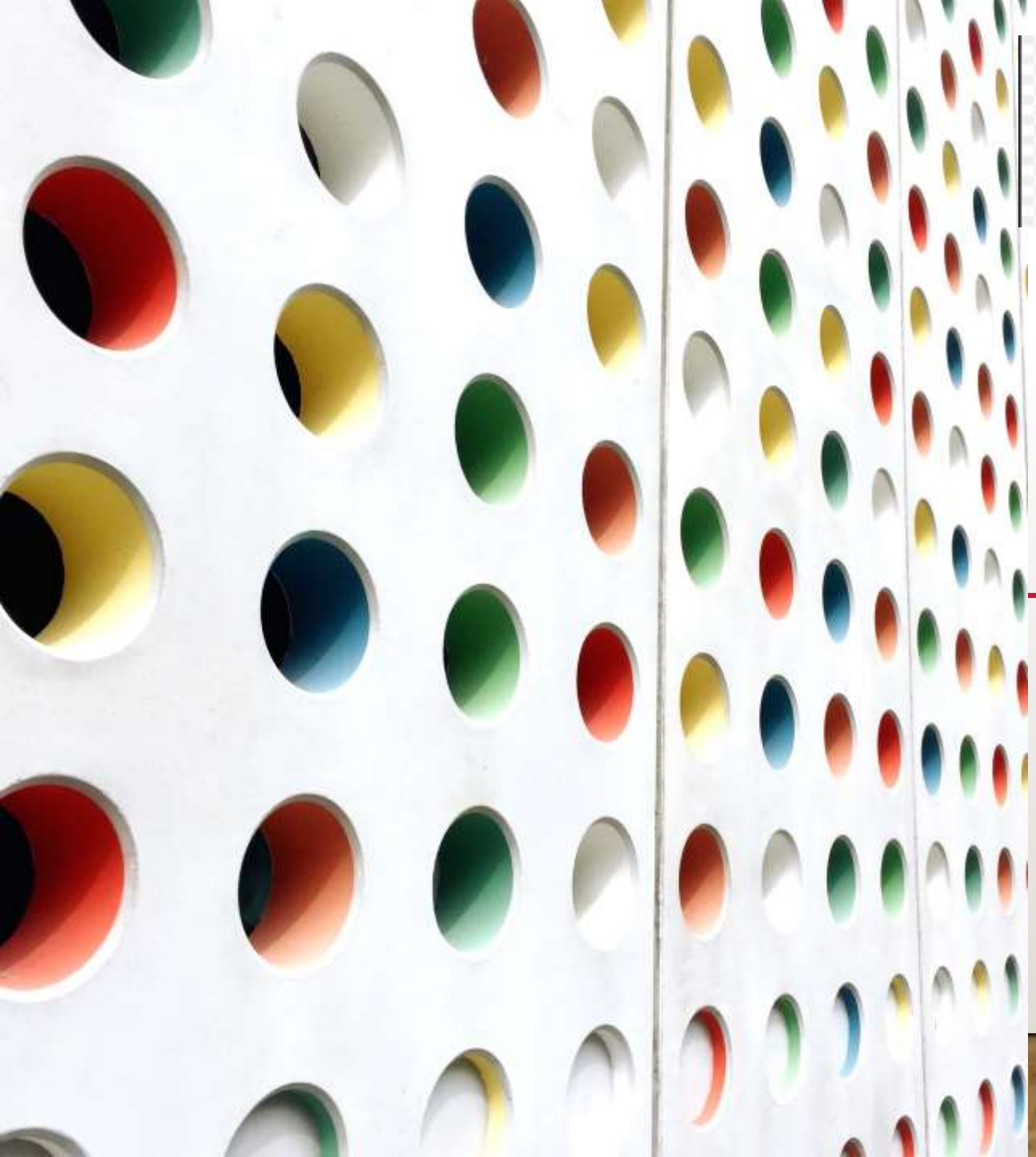


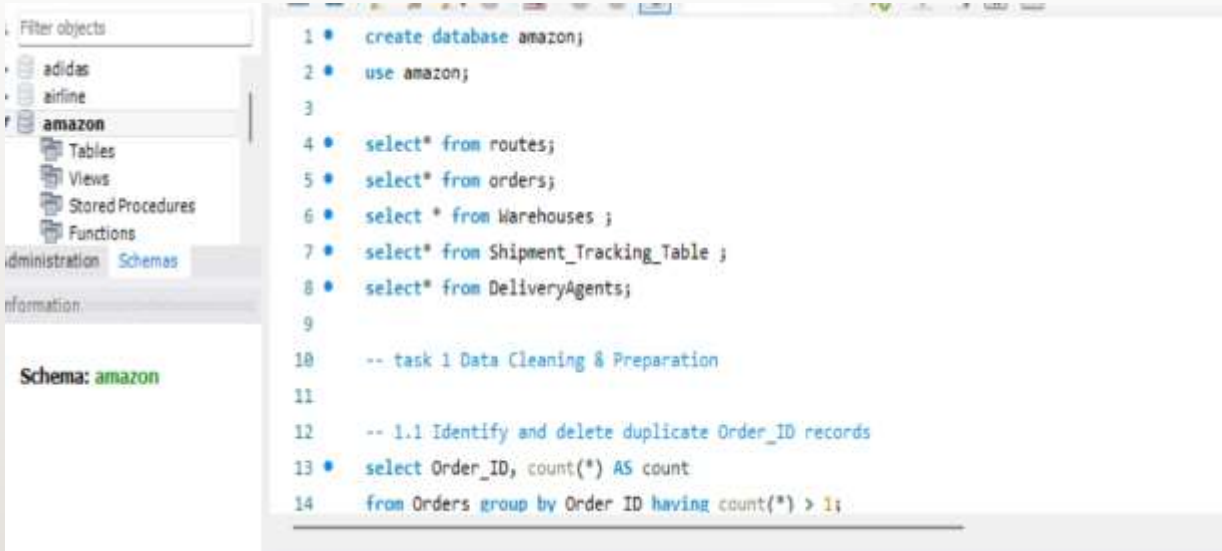
ISTICS IMIZATION FOR DELIVERY ROUTES – AMAZON

SQL-BASED ANALYTICS PROJECT

Rashmita Pradhan

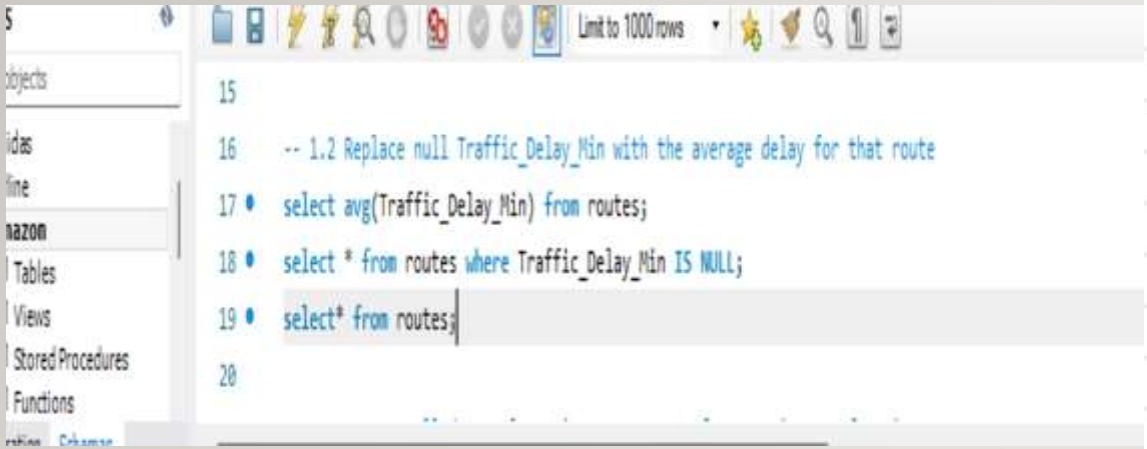
TASK 1: DATA CLEANING & PREPARATION

- 1) Identify and delete duplicate Order_ID records



Order_ID	Customer_ID	Warehouse_ID	Route_ID	Order_Date	Expected_Delivery_Date	Actual_Delivery_Date	Delivery_Status	Delivery_Delay_Days
O0001	C0080	W004	R005	2025-06-08	2025-06-11	2025-06-11	On Time	NULL
O0002	C0566	W009	R004	2025-06-22	2025-06-27	2025-06-29	Delayed	NULL
O0003	C0899	W008	R013	2025-06-04	2025-06-09	2025-06-11	Delayed	NULL
O0004	C0311	W001	R005	2025-06-28	2025-07-05	2025-07-07	Delayed	NULL
O0005	C0304	W006	R012	2025-06-04	2025-06-09	2025-06-10	Delayed	NULL
O0006	C0719	W004	R002	2025-07-17	2025-07-19	2025-07-19	On Time	NULL
O0007	C0652	W001	R020	2025-06-14	2025-06-17	2025-06-18	Delayed	NULL
O0008	C0111	W006	R008	2025-06-23	2025-06-30	2025-07-02	Delayed	NULL
O0009	C0138	W010	R011	2025-07-11	2025-07-16	2025-07-18	Delayed	NULL
O0010	C0096	W009	R017	2025-06-14	2025-06-21	2025-06-21	On Time	NULL
O0011	C0422	W008	R002	2025-05-26	2025-06-02	2025-06-02	On Time	NULL
O0012	C0838	W009	R014	2025-07-16	2025-07-20	2025-07-21	Delayed	NULL
O0013	C0543	W007	R002	2025-07-16	2025-07-18	2025-07-20	Delayed	NULL
O0014	C0385	W001	R007	2025-07-07	2025-07-11	2025-07-11	On Time	NULL
O0015	C0108	W001	R006	2025-06-18	2025-06-24	2025-06-24	On Time	NULL
O0016	C0077	W003	R005	2025-06-26	2025-06-29	2025-07-01	Delayed	NULL
O0017	C0703	W006	R014	2025-07-11	2025-07-17	2025-07-17	On Time	NULL

2) Replace null Traffic_Delay_Min with the average delay for that route



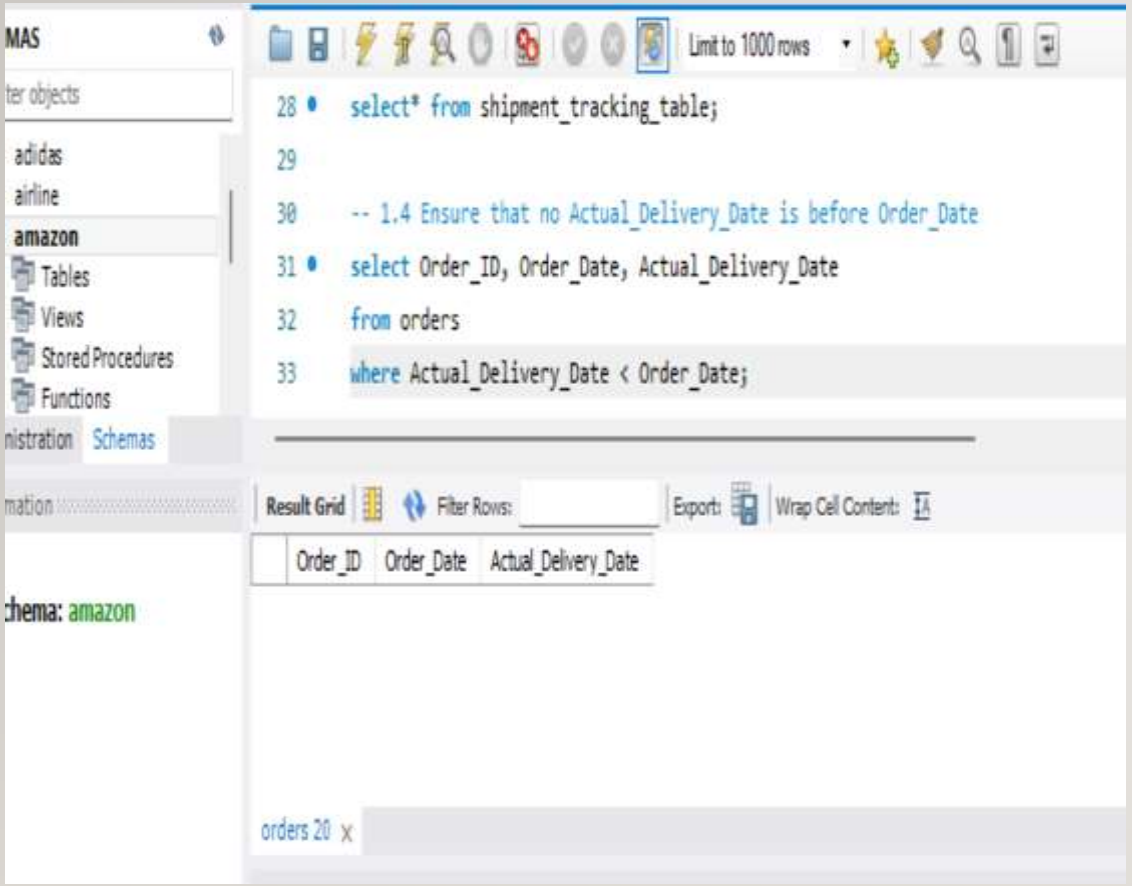
Route_ID	Start_Location	End_Location	Distance_KM	Average_Travel_Time_Min	Traffic_Delay_Min	Efficiency_Ratio
R001	New Aimeechester	Sabrinaland	130.1	212	36	NULL
R002	South Brianchester	Huffmanmouth	191.9	89	22	NULL
R003	East Amy	South Katherinemouth	110.5	190	11	NULL
R004	Ramosside	Stevenview	136.9	32	3	NULL
R005	New Jessica	Davismouth	71.4	191	26	NULL
R006	Lake Robinhaven	East Christopherville	29.4	183	42	NULL
R007	Aaronhaven	Jeanneberg	74.3	263	51	NULL
R008	South Davidborough	Carsonberg	52.4	215	34	NULL
R009	Bellfurt	Lake Marvin	126.4	209	46	NULL
R010	North Aliceside	Lake Jennifer	115	172	40	NULL
R011	Lowemouth	Samanthaborough	170.1	178	25	NULL
R012	West Calvinview	Port Audreyberg	99.4	110	29	NULL
R013	South Johnville	Port Scott	37.3	40	38	NULL
R014	North Edward	Matthewport	125.3	55	58	NULL
R015	South Jenniferfurt	Lindsayton	34.6	115	29	NULL
R016	Lake Kathrynchester	Port Michael	25.1	166	8	NULL
R017	Port Tammybury	Abbottside	24.6	75	8	NULL

3) Convert all date columns into YYYY-MM-DD format

```
update orders
set order_date = date(order_date),
expected_delivery_date = date(expected_delivery_date), actual_delivery_date = date(actual_delivery_date);
update shipment_tracking_table
set Checkpoint_Time = date(Checkpoint_Time);
select* from shipment_tracking_table;
```

Shipment_ID	Order_ID	Checkpoint	Checkpoint_Time	Delay_Reason
SO00011	O0001	Checkpoint 1	2025-06-09	Weather
SO00012	O0001	Checkpoint 2	2025-06-09	Sorting Delay
SO00021	O0002	Checkpoint 1	2025-06-22	Sorting Delay
SO00022	O0002	Checkpoint 2	2025-06-22	None
SO00023	O0002	Checkpoint 3	2025-06-23	Sorting Delay
SO00024	O0002	Checkpoint 4	2025-06-23	Weather
SO00031	O0003	Checkpoint 1	2025-06-04	Weather
SO00032	O0003	Checkpoint 2	2025-06-05	Traffic
SO00033	O0003	Checkpoint 3	2025-06-05	Traffic
SO00034	O0003	Checkpoint 4	2025-06-05	Weather
SO00041	O0004	Checkpoint 1	2025-06-28	Traffic
SO00042	O0004	Checkpoint 2	2025-06-29	Weather
SO00043	O0004	Checkpoint 3	2025-06-29	Traffic
SO00044	O0004	Checkpoint 4	2025-06-29	None
SO00045	O0004	Checkpoint 5	2025-06-29	Traffic
SO00051	O0005	Checkpoint 1	2025-06-04	None
SO00052	O0005	Checkpoint 2	2025-06-04	Weather

4) Ensure that no Actual_Delivery_Date is before Order_Date



- **No Duplicate Values or Null Values found.**
- **The data types of date columns were updated.**

TASK 2: DELIVERY DELAY ANALYSIS

- 1) Calculate delivery delay (in days) for each order

```
use amazon;

-- task 2.1 Calculate delivery delay (in days) for each order

alter table orders add column Delivery_Delay_Days int;
Select *, datediff(Actual_Delivery_Date, Expected_Delivery_Date) as delay from orders;
```

Order_ID	Customer_ID	Warehouse_ID	Route_ID	Order_Date	Expected_Delivery_Date	Actual_Delivery_Date	Delivery_Status	Delivery_Delay_Days	delay
O0001	C0080	W004	R005	2025-06-08	2025-06-11	2025-06-11	On Time	NULL	0
O0002	C0566	W009	R004	2025-06-22	2025-06-27	2025-06-29	Delayed	NULL	2
O0003	C0899	W008	R013	2025-06-04	2025-06-09	2025-06-11	Delayed	NULL	2
O0004	C0311	W001	R005	2025-06-28	2025-07-05	2025-07-07	Delayed	NULL	2
O0005	C0304	W006	R012	2025-06-04	2025-06-09	2025-06-10	Delayed	NULL	1
O0006	C0719	W004	R002	2025-07-17	2025-07-19	2025-07-19	On Time	NULL	0
O0007	C0652	W001	R020	2025-06-14	2025-06-17	2025-06-18	Delayed	NULL	1
O0008	C0111	W006	R008	2025-06-23	2025-06-30	2025-07-02	Delayed	NULL	2
O0009	C0138	W010	R011	2025-07-11	2025-07-16	2025-07-18	Delayed	NULL	2
O0010	C0096	W009	R017	2025-06-14	2025-06-21	2025-06-21	On Time	NULL	0
O0011	C0422	W008	R002	2025-05-26	2025-06-02	2025-06-02	On Time	NULL	0
O0012	C0838	W009	R014	2025-07-16	2025-07-20	2025-07-21	Delayed	NULL	1
O0013	C0543	W007	R002	2025-07-16	2025-07-18	2025-07-20	Delayed	NULL	2
O0014	C0385	W001	R007	2025-07-07	2025-07-11	2025-07-11	On Time	NULL	0

2) Top 10 delayed routes based on average delay days

```
--- task 2.2 Top 10 delayed routes based on average delay days

Select Route_id, avg(datediff(Actual_Delivery_Date, Expected_Delivery_Date)) as avg_delay
from orders group by route_id order by avg_delay desc Limit 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Route_id	avg_delay
R012	1.0833
R020	1.0588
R017	1.0000
R015	1.0000
R005	0.8000

Result 3

Output

Route_id	avg_delay
R012	1.0833
R020	1.0588
R017	1.0000
R015	1.0000
R005	0.8000
R004	0.8000
R013	0.7647
R011	0.7000
R010	0.7000
R007	0.6875

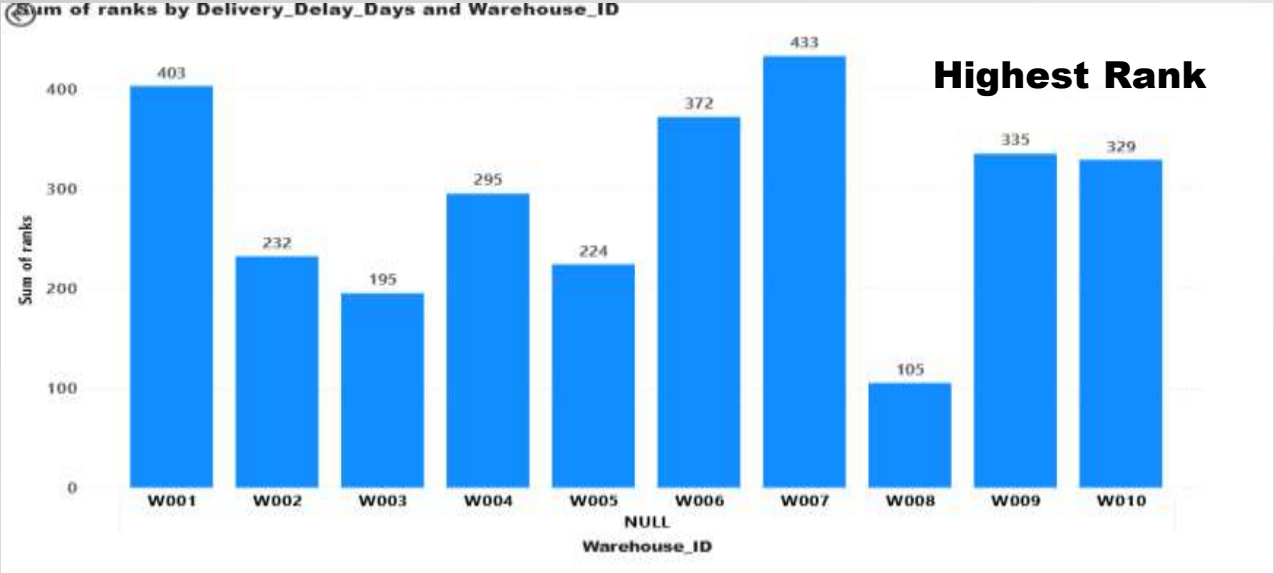
3) Use window functions to rank all orders by delay within each warehouse

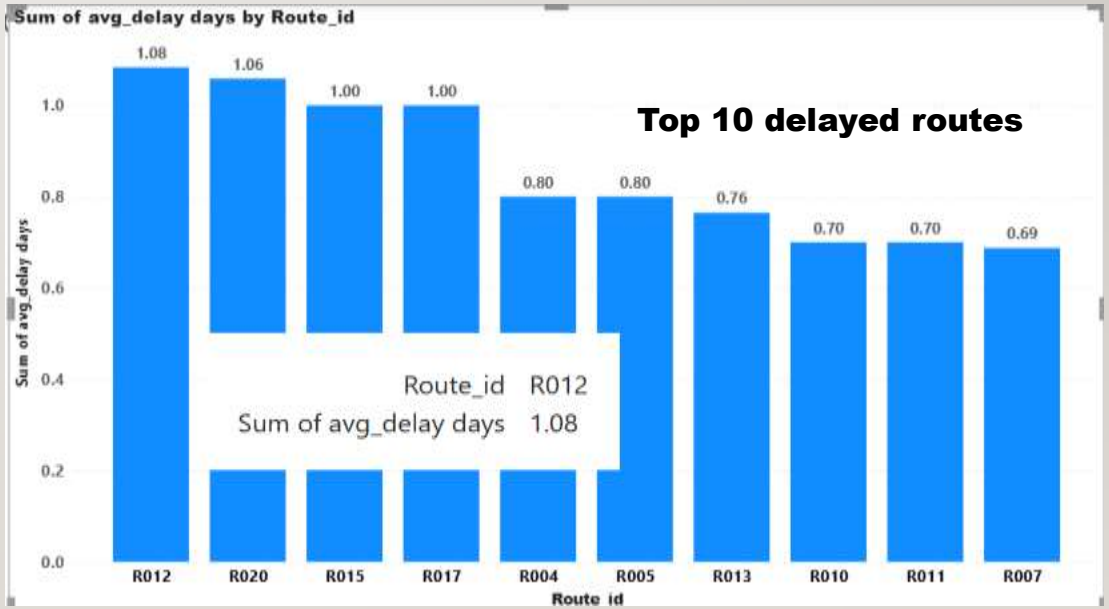
```sql
-- task 2.3  rank all orders by delay within each warehouse

select *, rank() over (partition by Warehouse_Id order by datediff(Actual_Delivery_Date, Expected_Delivery_Date) desc) as ranks
from orders order by ranks desc;
```

Order_ID	Customer_ID	Warehouse_ID	Route_ID	Order_Date	Expected_Delivery_Date	Actual_Delivery_Date	Delivery_Status	Delivery_Delay_Days	ranks
O0062	C0292	W010	R003	2025-07-09	2025-07-13	2025-07-13	On Time	NULL	21
O0136	C0770	W010	R017	2025-07-12	2025-07-18	2025-07-18	On Time	NULL	21
O0072	C0973	W010	R019	2025-06-17	2025-06-23	2025-06-23	On Time	NULL	21
O0216	C0069	W010	R013	2025-06-10	2025-06-15	2025-06-15	On Time	NULL	21
O0030	C0426	W010	R010	2025-06-26	2025-07-02	2025-07-02	On Time	NULL	21
O0039	C0210	W010	R001	2025-07-10	2025-07-13	2025-07-13	On Time	NULL	21
O0038	C0382	W010	R013	2025-05-23	2025-05-26	2025-05-26	On Time	NULL	21
O0095	C0784	W010	R007	2025-05-23	2025-05-29	2025-05-29	On Time	NULL	21
O0196	C0040	W010	R006	2025-07-11	2025-07-18	2025-07-18	On Time	NULL	21
O0181	C0632	W010	R006	2025-05-30	2025-06-02	2025-06-02	On Time	NULL	21
O0070	C0160	W001	R008	2025-06-08	2025-06-13	2025-06-13	On Time	NULL	19
O0210	C0308	W001	R013	2025-07-07	2025-07-11	2025-07-11	On Time	NULL	19
O0289	C0874	W001	R011	2025-05-26	2025-05-29	2025-05-29	On Time	NULL	19
O0215	C0159	W001	R016	2025-07-16	2025-07-21	2025-07-21	On Time	NULL	19
O0290	C0603	W001	R004	2025-05-29	2025-06-04	2025-06-04	On Time	NULL	19

Task 2 output and insights:-

- The delivery delay was calculate.
- The for top 10 routes based on avg delay are displayed average delay days.
- Used Windows function to rank all orders by delay within each warehouse.
- Warehouse_id W007 has highest rank.
- Top 10 delayed routes is shown as per avg delay days.





TASK 3: ROUTE OPTIMIZATION INSIGHTS

- 1) * Average delivery time

```
-- 1.1 Average delivery time (in days)
SELECT r.Route_ID, r.Start_Location, r.End_Location,
ROUND(AVG(DATEDIFF(o.Actual_Delivery_Date, o.Order_Date)),2) AS Avg_Delivery_Time_Days
FROM routes r
INNER JOIN orders o ON r.Route_ID = o.Route_ID
GROUP BY 1 , 2 , 3;
```

Route_ID	Start_Location	End_Location	Avg_Delivery_Time_Days
R005	New Jessica	Davismouth	5.00
R004	Ramosside	Stevenview	5.10
R013	South Johnville	Port Scott	4.41
R012	West Calvinview	Port Audreyberg	4.83
R002	Brianchester	Huffmanmouth	4.73
R020	Juliestad	South Jonathanshire	5.88
R008	South Davidborough	Carsonberg	5.44
R011	Lowemouth	Samanthaborough	5.55
R017	Port Tammybury	Abbottside	6.29
R014	North Edward	Matthewport	5.62
R007	Aaronhaven	Jeanneberg	5.75
R006	Lake Robinhaven	East Christopherville	4.82
R003	East Amy	South Katherinemouth	4.50
R015	South Jenniferfurt	Lindsayton	6.33
R009	Bellfurt	Lake Marvin	4.88
R010	North Aliceside	Lake Jennifer	5.25
R019	Lake Oscarfurt	New Ivanshire	5.62

* Average traffic delay

```sql
-- 1.2 Average traffic delay
SELECT Route_ID, Start_Location, End_Location,
ROUND(AVG(Traffic_Delay_Min),2) AS Avg_Traffic_Delay_Min
FROM routes
GROUP BY 1,2,3;
```

Route_ID	Start_Location	End_Location	Avg_Traffic_Delay_Min
R001	New Aimeechester	Sabrinaland	36.00
R002	South Brianchester	Huffmanmouth	22.00
R003	East Amy	South Katherinemouth	11.00
R004	Ramosside	Stevenview	3.00
R005	New Jessica	Davismouth	26.00
R006	Lake Robinhaven	East Christopherville	42.00
R007	Aaronhaven	Jeanneberg	51.00
R008	South Davidborough	Carsonberg	34.00
R009	Bellfurt	Lake Marvin	46.00
R010	North Aliceside	Lake Jennifer	40.00
R011	Lowemouth	Samanthaborough	25.00
R012	West Calvinview	Port Audreyberg	29.00
R013	South Johnville	Port Scott	38.00
R014	North Edward	Matthewport	58.00
R015	South Jenniferfurt	Lindsayton	29.00
R016	Lake Kathrynchester	Port Michael	8.00
R017	Port Tammybury	Abbottside	8.00

* Distance-to-time efficiency ratio: Distance_KM / Average_Travel_Time_Min

```
-- 1.3 Distance-to-time efficiency ratio: Distance_KM / Average_Travel_Time_Min
ALTER TABLE routes ADD COLUMN Efficiency_Ratio DECIMAL(10,4);
Select route_id, round(Distance_KM / Average_Travel_Time_Min,2) as effeciency_ratio
from routes order by effeciency_ratio;
select* from routes;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

route_id	effeciency_ratio
R016	0.15
R006	0.16
R008	0.24
R007	0.28
R015	0.3

Result 14

Output

route_id	effeciency_ratio
R016	0.15
R006	0.16
R008	0.24
R007	0.28
R015	0.3
R017	0.33
R005	0.37
R018	0.54
R020	0.55
R003	0.58
R009	0.6
R001	0.61
R010	0.67
R012	0.9
R013	0.93
R011	0.96
R002	2.16

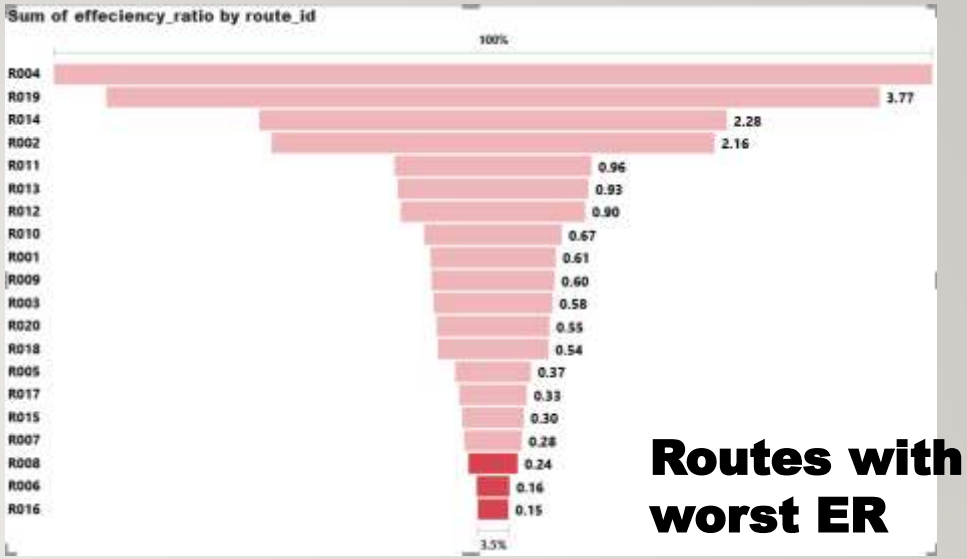
0.16

2) Identify 3 routes with the worst efficiency ratio.

```
Select route_id, (Distance_KM / Average_Travel_Time_Min) as effeciency_ratio
from routes
order by effeciency_ratio
limit 3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

route_id	effeciency_ratio
R016	0.15120481927710844
R006	0.16065573770491803
R008	0.24372093023255814



Routes with worst ER

3) Find routes with >20% delayed shipments

```
-- task 3 routes with >20% delayed shipments
SELECT  Route_ID, COUNT(*) AS Total_Orders, SUM(CASE
WHEN DATEDIFF(Actual_Delivery_Date, Expected_Delivery_Date) > 0 THEN 1 ELSE 0 END) AS Delayed_Orders,
(SUM(CASE WHEN DATEDIFF(Actual_Delivery_Date, Expected_Delivery_Date) > 0 THEN 1 ELSE 0 END) * 100.0 / COUNT(*)) AS Delay_Percentage
FROM Orders GROUP BY Route_ID HAVING Delay_Percentage > 20 order by Delay_Percentage ASC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Route_ID	Total_Orders	Delayed_Orders	Delay_Percentage
R016	12	3	25.00000
R019	13	4	30.76923
R008	16	5	31.25000
R002	22	7	31.81818
R006	11	4	36.36364

Result 23 ×

Output

Route_ID	Total_Orders	Delayed_Orders	Delay_Percentage
R016	12	3	25.00000
R019	13	4	30.76923
R008	16	5	31.25000
R002	22	7	31.81818
R006	11	4	36.36364
R009	16	6	37.50000
R014	13	5	38.46154
R003	10	4	40.00000
R001	15	6	40.00000
R018	15	6	40.00000
R007	16	7	43.75000
R011	20	9	45.00000
R010	20	9	45.00000
R005	15	7	46.66667
R004	20	10	50.00000
R015	6	3	50.00000
R013	17	9	52.94118

4) Recommend potential routes for optimization

```sql
-- Route Optimization

SELECT r.Route_ID,ROUND(r.Distance_KM / r.Average_Travel_Time_Min, 2) AS Efficiency_Ratio, COUNT(o.Order_ID) AS Total_Orders,
SUM(CASE WHEN DATEDIFF(o.Actual_Delivery_Date, o.Expected_Delivery_Date) > 0
THEN 1 ELSE 0 END) AS Delayed_Orders,ROUND(SUM(CASE WHEN DATEDIFF(o.Actual_Delivery_Date, o.Expected_Delivery_Date) > 0
THEN 1 ELSE 0 END) * 100.0 / COUNT(o.Order_ID), 2) AS Delay_Percentage FROM routes r JOIN orders o ON r.Route_ID = o.Route_ID
GROUP BY r.Route_ID, r.Distance_KM, r.Average_Travel_Time_Min HAVING Delay_Percentage > 20   -- High delay
AND Efficiency_Ratio < (SELECT AVG(Distance_KM / Average_Travel_Time_Min) FROM routes) ORDER BY Delay_Percentage DESC;
```

Route_ID	Efficiency_Ratio	Total_Orders	Delayed_Orders	Delay_Percentage
R020	55	17	12	70.59
R012	0.9	12	8	66.67
R017	0.33	14	8	57.14
R013	0.93	17	9	52.94
R015	0.3	6	3	50.00
R005	0.37	15	7	46.67
R011	0.96	20	9	45.00
R010	0.67	20	9	45.00
R007	0.28	16	7	43.75
R003	0.58	10	4	40.00

Task 3 output and insights:-

- **The Average delivery time (in days).Average traffic delay and efficiency ratio was calculated.**
- **3 routes with the worst efficiency ratio is shown.**
- **Windows functions is used to rank all orders by delay within each route.**
- **Routes with high delay percentage and low efficiency ratio are recommended for optimization as they indicate poor delivery performance.**

.

TASK 4: WAREHOUSE PERFORMANCE

```sql
use amazon;
-- task 1 Find the top 3 warehouses with the highest average processing time
SELECT Warehouse_ID, Location, ROUND(AVG(Processing_Time_Min), 2) AS Avg_Processing_Time_Min
FROM warehouses GROUP BY Warehouse_ID, Location ORDER BY Avg_Processing_Time_Min DESC LIMIT 3;
```

Warehouse_ID	Location	Avg_Processing_Time_Min
W010	Davidshire	171.00
W009	New Crystalborough	170.00
W007	Port Juliachester	166.00

```sql
-- task 2 Calculate total vs. delayed shipments for each warehouse.
SELECT  Warehouse_ID, COUNT(*) AS Total_Orders,SUM(
CASE
WHEN DATEDIFF(Actual_Delivery_Date, Expected_Delivery_Date) > 0 THEN 1 ELSE 0
END) AS Delayed_Orders FROM Orders GROUP BY Warehouse_ID;
```

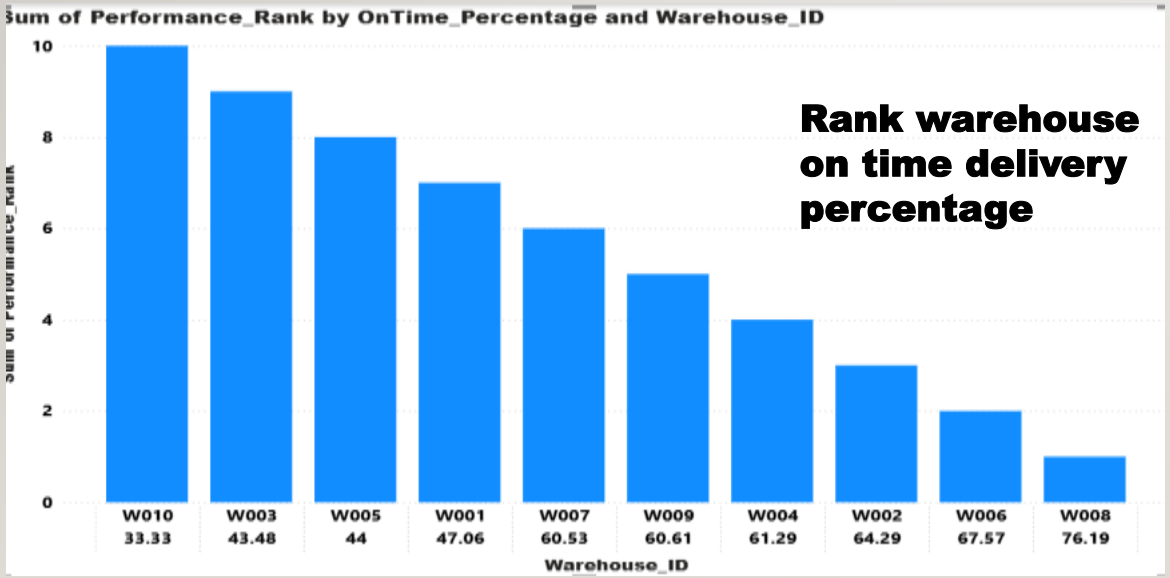
Warehouse_ID	Total_Orders	Delayed_Orders
W009	33	13
W008	21	5
W001	34	18
W006	37	12
W010	30	20
W007	38	15
W003	23	13
W005	25	14
W002	28	10

```sql
-- task 3: Use CTEs to find bottleneck warehouses where processing time > global average
WITH AvgProcessing AS (SELECT AVG(Processing_Time_Min) AS Global_Avg_Processing_Time FROM warehouses)
SELECT w.Warehouse_ID, w.Location, w.Processing_Time_Min, a.Global_Avg_Processing_Time,
CASE WHEN w.Processing_Time_Min > a.Global_Avg_Processing_Time THEN 'Bottleneck' ELSE 'Normal'
END AS Warehouse_Status FROM warehouses w JOIN AvgProcessing a ORDER BY w.Processing_Time_Min DESC;
```

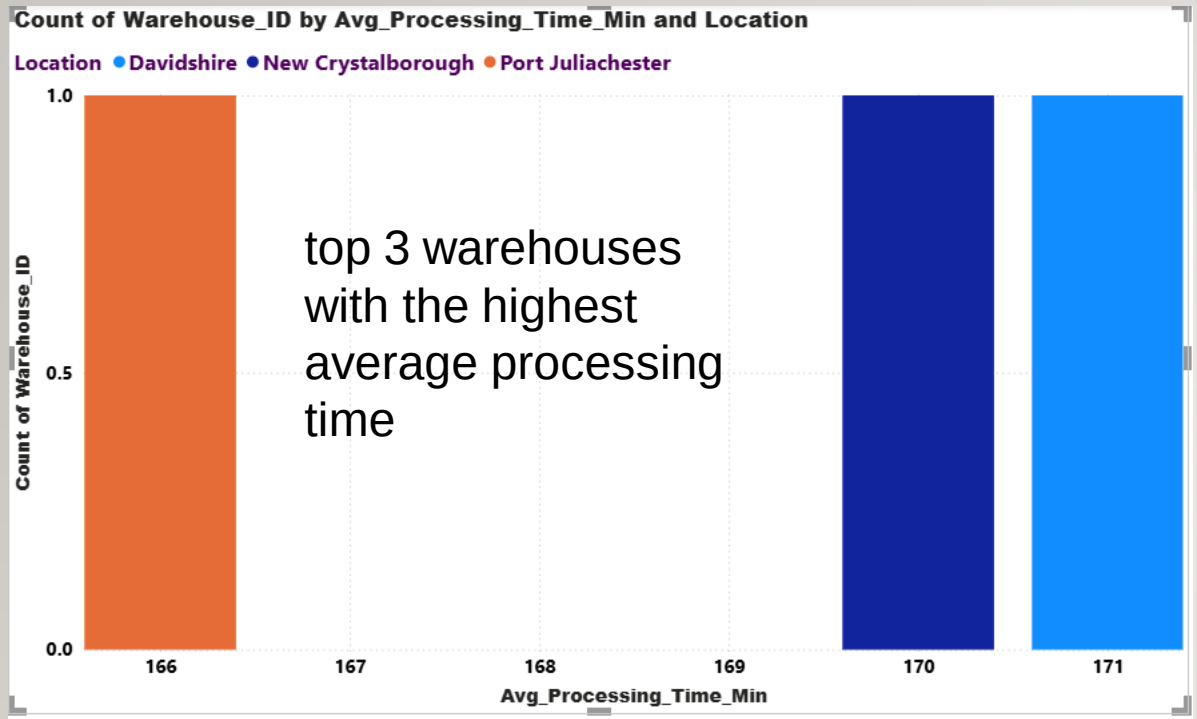
Warehouse_ID	Location	Processing_Time_Min	Global_Avg_Processing_Time	Warehouse_Status
W010	Davidshire	171	110.1000	Bottleneck
W009	New Crystalborough	170	110.1000	Bottleneck
W007	Port Juliachester	166	110.1000	Bottleneck
W008	Andersontown	161	110.1000	Bottleneck
W001	Zacharyland	142	110.1000	Bottleneck
W004	Barnettville	84	110.1000	Normal
W005	Lake Patriciaside	78	110.1000	Normal
W002	Smithport	73	110.1000	Normal
W003	Lake Cody	29	110.1000	Normal
W006	Petershaven	27	110.1000	Normal

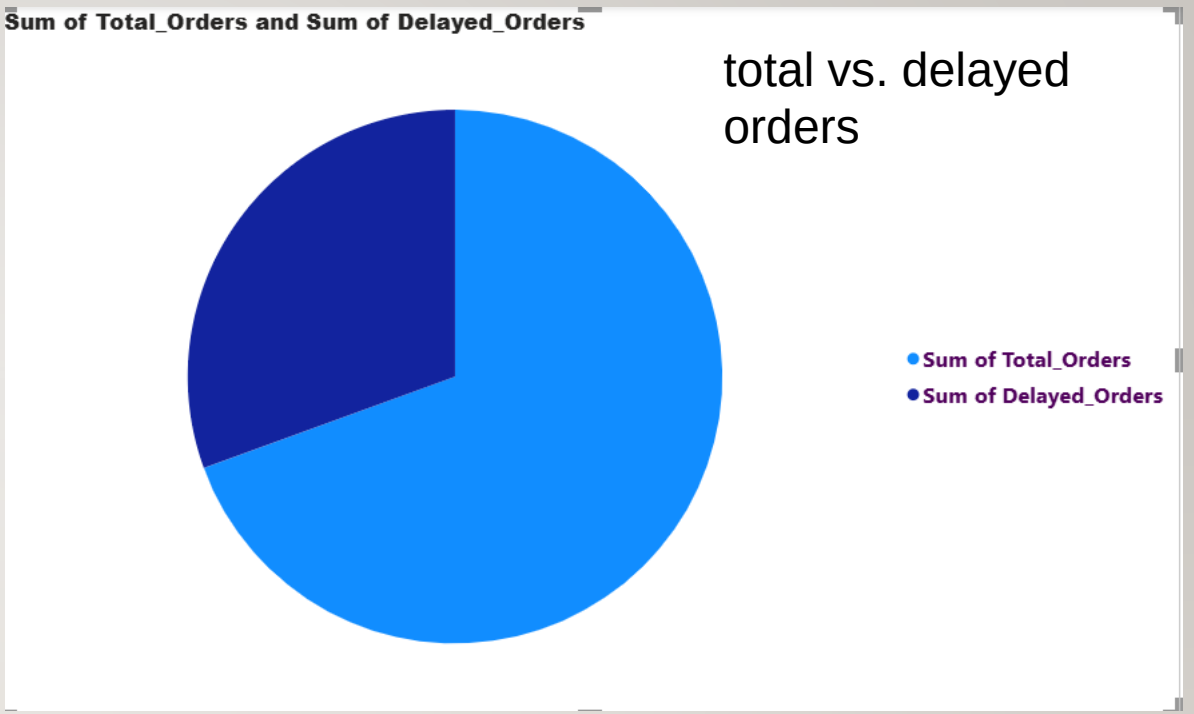
```sql
-- task 4: Rank warehouses based on on-time delivery percentage
SELECT w.Warehouse_ID, w.Location,ROUND((SUM(CASE WHEN o.Delivery_Status = 'On Time' THEN 1 ELSE 0 END) * 100) / COUNT(*), 2)
AS OnTime_Percentage, RANK() OVER (ORDER BY (SUM(CASE WHEN o.Delivery_Status = 'On Time' THEN 1 ELSE 0 END) * 100 / COUNT(*)) DESC)
AS Performance_Rank FROM warehouses w JOIN orders o ON w.Warehouse_ID = o.Warehouse_ID GROUP BY w.Warehouse_ID, w.Location
ORDER BY OnTime_Percentage DESC;
```

Warehouse_ID	Location	OnTime_Percentage	Performance_Rank
W008	Andersontown	76.19	1
W006	tershaven	67.57	2
W002	Smithport	64.29	3
W004	Barnettville	61.29	4
W009	New Crystalborough	60.61	5
W007	Port Juliachester	60.53	6
W001	Zacharyland	47.06	7
W005	Lake Patriciaside	44.00	8
W003	Lake Cody	43.48	9
W010	Davidshire	33.33	10



Rank warehouse on time delivery percentage





Task 4 output and insights:-

- **Total vs. delayed orders for each warehouse are calculated.**
- **Some warehouses consistently show high processing time and high delay %.**
- **Top 3 warehouses with the highest average processing time are Davidshire, New crystalborough and Port juliachester.**
- **Bottleneck warehouses are the primary contributors to delivery inefficiency.**

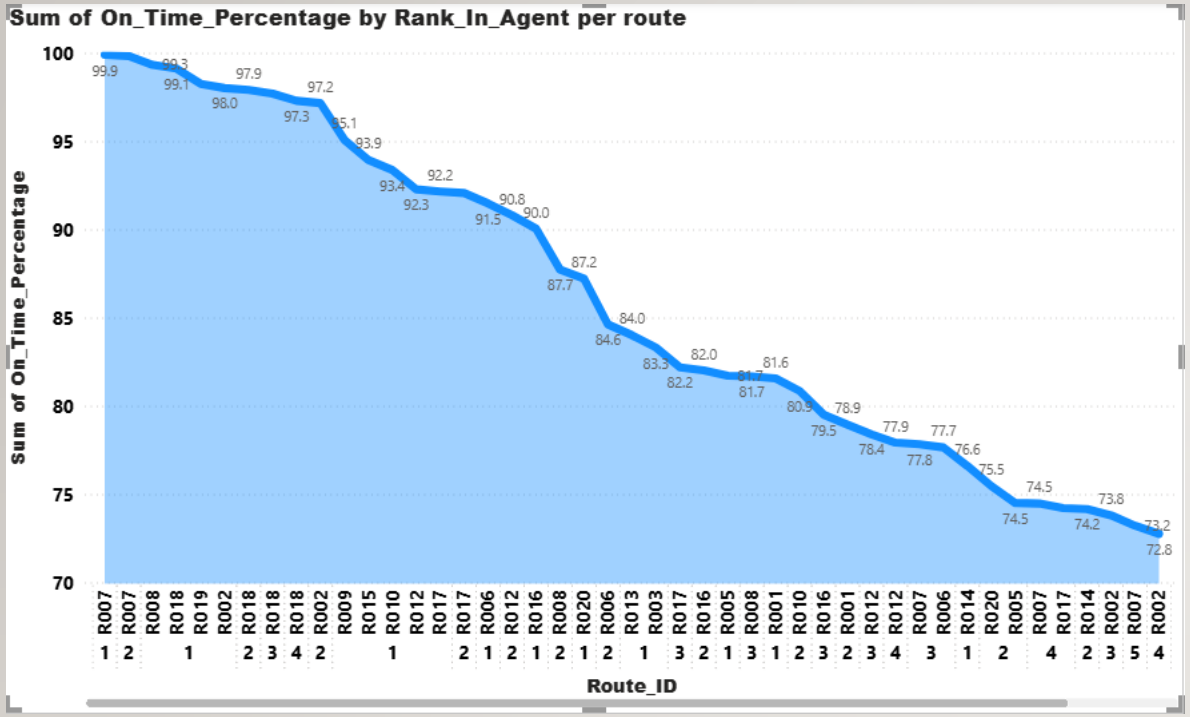
TASK 5: DELIVERY AGENT PERFORMANCE

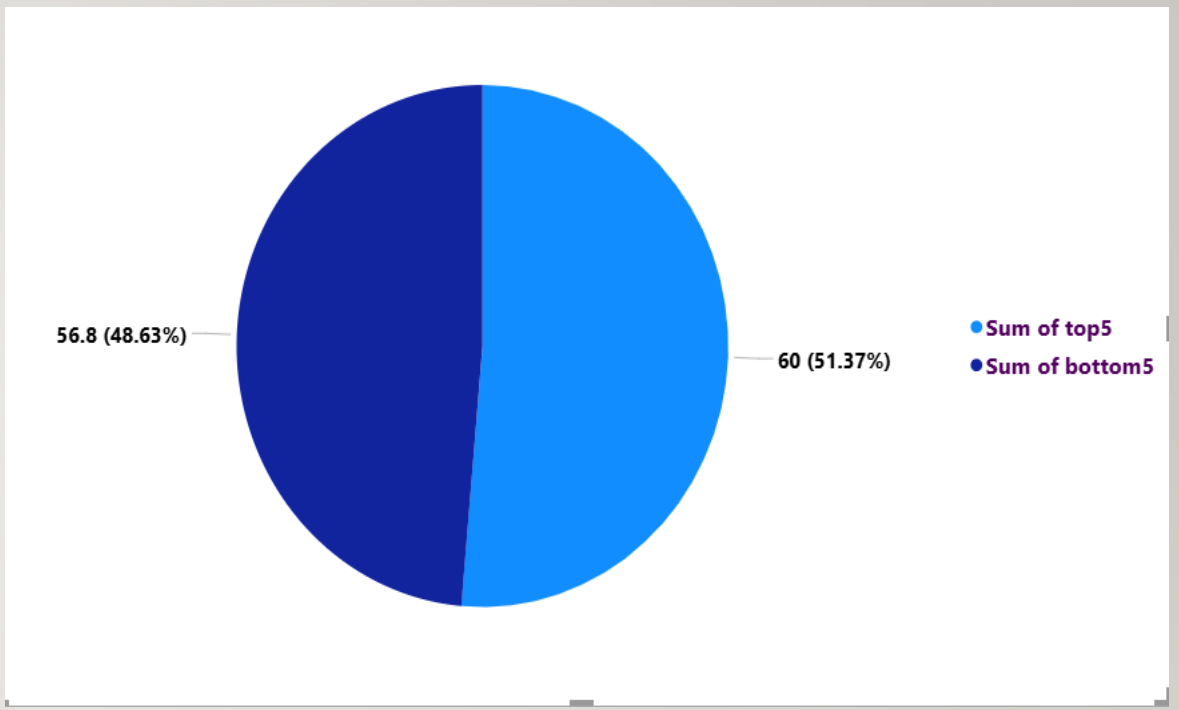
```sql
use amazon;
-- task 1 • Rank agents (per route) by on-time delivery percentage
SELECT Agent_ID, Route_ID, On_Time_Percentage,
RANK() OVER (PARTITION BY Route_ID ORDER BY On_Time_Percentage DESC ) AS Rank_In_Agent
FROM deliveryagents ORDER BY Route_ID, Rank_In_Agent;

-- task 2 Find agents with on-time % < 80%
SELECT * from deliveryagents WHERE On_Time_Percentage < 80 ORDER BY On_Time_Percentage;

-- task 3 Compare average speed of top 5 vs bottom 5 agents using subqueries
Select avg(t.avg_speed) as top5, avg(b.avg_speed) as
bottom5 from(Select d.agent_id, d.Avg_Speed_KM_HR as avg_speed from deliveryagents d
order by On_Time_Percentage desc Limit 5) t
cross join (Select d.agent_id, d.Avg_Speed_KM_HR as avg_speed from deliveryagents d order by On_Time_Percentage asc Limit 5) b;
```

Agent_ID	Route_ID	On_Time_Percentage	Rank_In_Agent
A006	R001	81.56	1
A037	R001	78.94	2
A033	R002	98.01	1
A024	R002	97.16	2
A027	R002	73.8	3
A050	R002	72.75	4
A011	R002	71.5	5
A015	R003	83.31	1
A029	R005	81.71	1
A035	R005	74.5	2

Agent_ID	Route_ID	Shift_Hours	Avg_Speed_KM_HR	On_Time_Percentage
A010	R019	9	76	70.45
A011	R002	4	59	71.5
A038	R018	9	55	71.8
A008	R018	9	32	72.39
A039	R016	9	62	72.6
A050	R002	8	69	72.75
A018	R007	5	75	73.21
A027	R002	7	69	73.8
A003	R014	5	72	74.15
A023	R017	8	59	74.21

top5	bottom5
60.0000	56.8000





Task 5 output and insights:-

- Agents were ranked within each route based on their on-time delivery percentage.
- Agents with on-time delivery below 80% were taken out.
- Average speed of top 5 vs bottom 5 agents .is compared.
- Faster agents tend to deliver more reliably and on time.

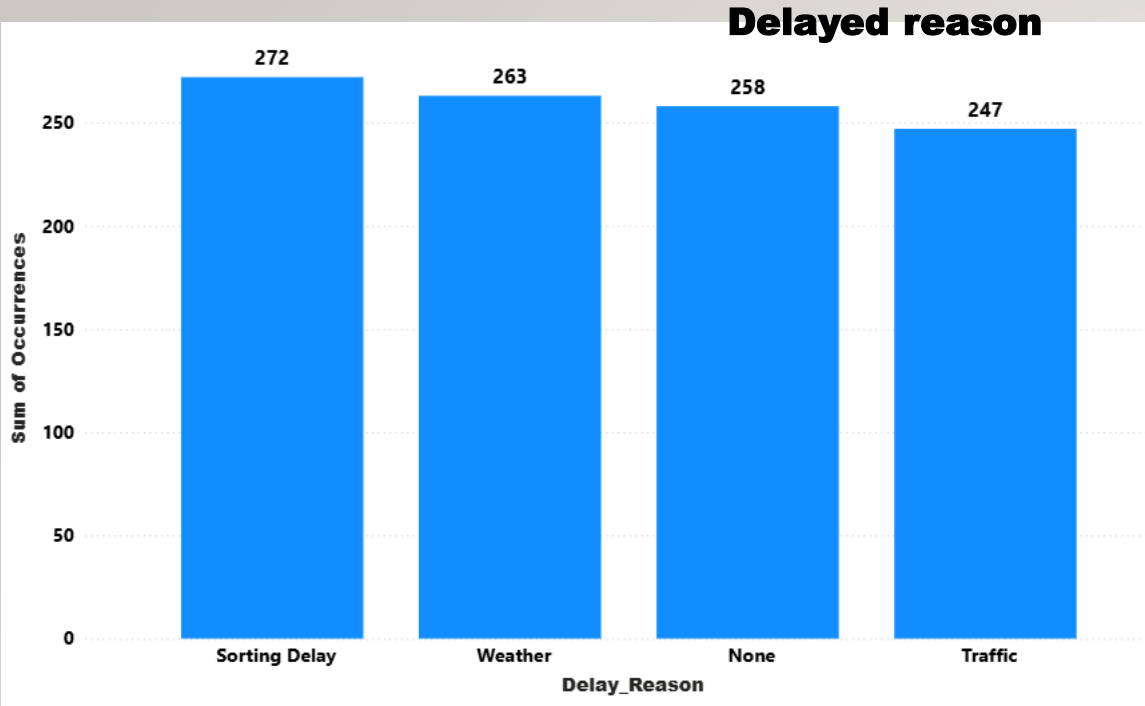
TASK 6: SHIPMENT TRACKING ANALYTICS

```sql
USE AMAZON;

-- task 1 For each order, list the last checkpoint and time
SELECT  Order_ID, MAX(Checkpoint) AS Last_Checkpoint, MAX(Checkpoint_Time) AS Last_Checkpoint_Time
FROM shipment_tracking_table GROUP BY Order_ID ORDER BY Order_ID;

-- task 2 Find the most common delay reasons (excluding None)
SELECT Delay_Reason, COUNT(*) AS Occurrences FROM shipment_tracking_table
WHERE Delay_Reason IS NOT NULL GROUP BY Delay_Reason ORDER BY Occurrences DESC;

-- task 3   Identify orders with >2 delayed checkpoints
SELECT Order_ID,COUNT(Checkpoint) AS Delayed_Checkpoint_Count FROM shipment_tracking_table
WHERE Delay_Reason= 'None'GROUP BY Order_ID HAVING COUNT(Checkpoint) > 2 ORDER BY Delayed_Checkpoint_Count DESC;
```

Order_ID	Last_Checkpoint	Last_Checkpoint_Time
O0001	Checkpoint 2	2025-06-09
O0002	Checkpoint 4	2025-06-23
O0003	Checkpoint 4	2025-06-05
O0004	Checkpoint 5	2025-06-29
O0005	Checkpoint 5	2025-06-04
O0006	Checkpoint 2	2025-07-18
O0007	Checkpoint 3	2025-06-15
O0008	Checkpoint 2	2025-06-24
O0009	Checkpoint 3	2025-07-11
O0010	Checkpoint 4	2025-06-15
O0011	Checkpoint 5	2025-05-27
O0012	Checkpoint 3	2025-07-17
O0013	Checkpoint 2	2025-07-17
O0014	Checkpoint 3	2025-07-08
O0015	Checkpoint 5	2025-06-20

Delay_Reason	Occurrences
Sorting Delay	272
Weather	263
None	258
Traffic	247

Order_ID	Delayed_Checkpoint_Count
O0011	4
O0204	4
O0237	4
O0265	4
O0018	3
O0020	3
O0042	3
O0059	3
O0109	3
O0150	3
O0151	3
O0194	3
O0197	3
O0201	3
O0226	3



Task 6 output and insights:-

- **A small number of reasons contributed to most shipment delays, typically like Traffic congestion, Weather conditions, Operational or warehouse issues.**
- **Orders having more than two delayed checkpoints were taken out.**
- **Delay reason analysis helps target root causes, reducing future disruptions..**

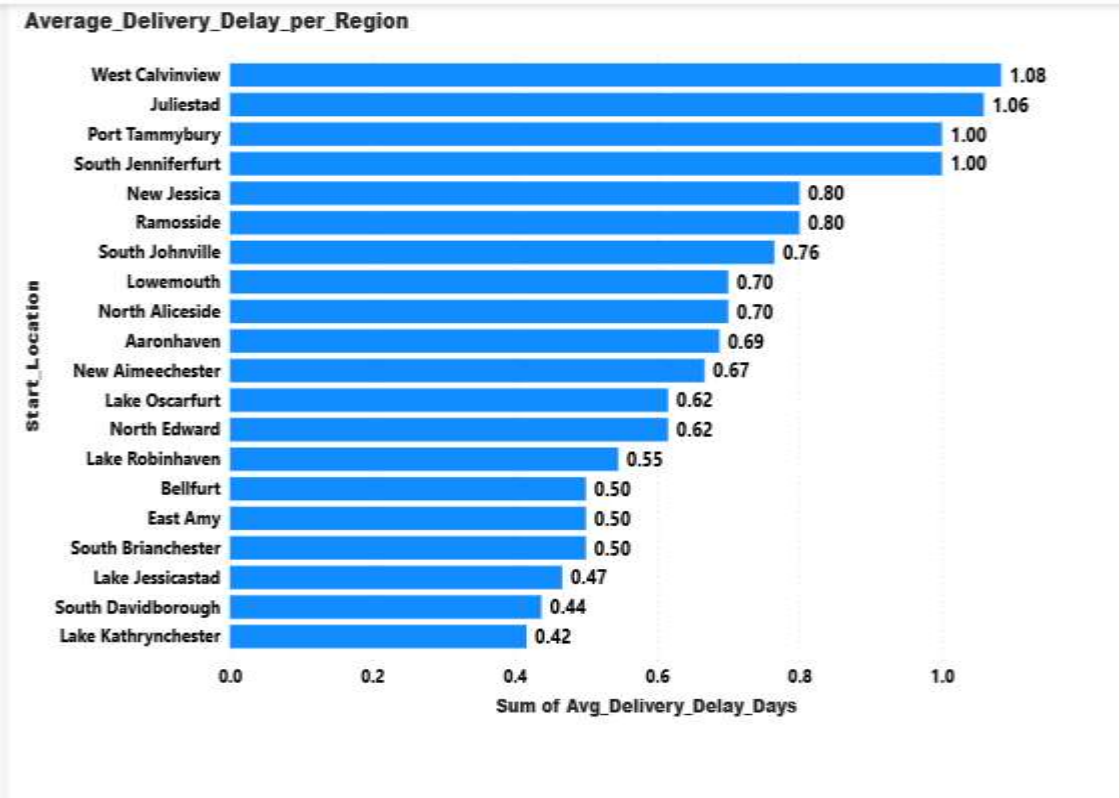
TASK 7: ADVANCED KPI REPORTING

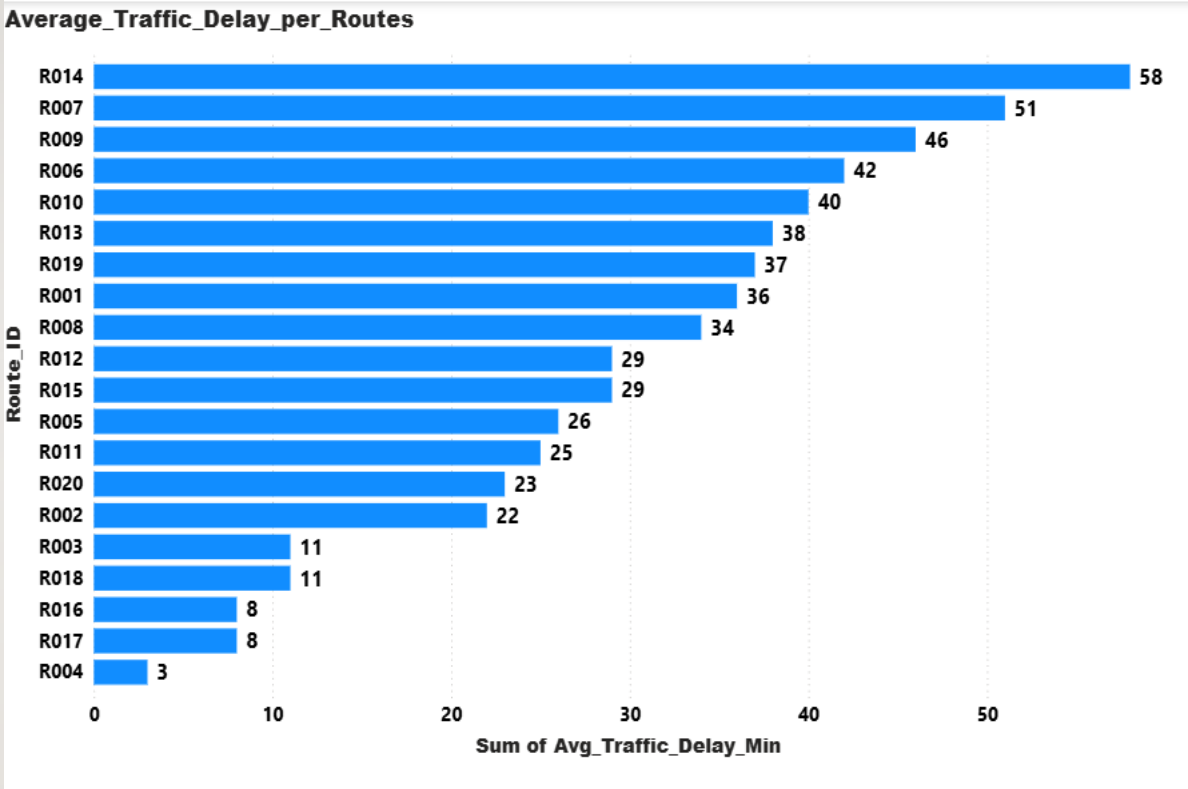
```sql
use amazon;

-- Task :- Calculate KPIs using SQL queries:
-- task 1.1 Average Delivery Delay per Region (Start_Location)
SELECT r.Start_Location, AVG(DATEDIFF(o.Actual_Delivery_Date, o.Expected_Delivery_Date)) AS Avg_Delivery_Delay_Days
FROM Orders o JOIN Routes r ON o.Route_ID = r.Route_ID GROUP BY r.Start_Location;


-- task 1.2 On-Time Delivery % = (Total On-Time Deliveries / Total Deliveries) * 100
SELECT COUNT(*) AS Total_Deliveries, SUM( CASE WHEN DATEDIFF(Actual_Delivery_Date, Expected_Delivery_Date) <= 0 THEN 1 ELSE 0
END) AS On_Time_Deliveries,(SUM(CASE WHEN DATEDIFF(Actual_Delivery_Date, Expected_Delivery_Date) <= 0 THEN 1 ELSE 0
END) * 100.0 / COUNT(*)) AS On_Time_Delivery_Percentage FROM Orders;


 -- task 1.3 Average Traffic Delay per Route
SELECT Route_ID, AVG(Traffic_Delay_Min) AS Avg_Traffic_Delay_Min FROM Routes GROUP BY Route_ID;
```

Start_Location	Avg_Delivery_Delay_Days
New Jessica	0.8000
Ramosside	0.8000
South Johnville	0.7647
West Calvinview	1.0833
South Brianchester	0.5000
Juliestad	1.0588
South Davidborough	0.4375
Lowemouth	0.7000
Port Tammybury	1.0000
North Edward	0.6154
Aaronhaven	0.6875
Lake Robinhaven	0.5455
East Amy	0.5000
South Jenniferfurt	1.0000
Bellfurt	0.5000

Total_Deliveries	On_Time_Deliveries	On_Time_Delivery_Percentage
300	168	56.00000

Route_ID	Avg_Traffic_Delay_Min
R001	36.0000
R002	22.0000
R003	11.0000
R004	3.0000
R005	26.0000
R006	42.0000
R007	51.0000
R008	34.0000
R009	46.0000
R010	40.0000
R011	25.0000
R012	29.0000
R013	38.0000
R014	58.0000
R015	29.0000





Task 7 output and insights:-

- **Average Delivery Delay per Region where west calvinview has avg delivery delay days.**
- **Average Traffic Delay per Route are shown where route R014 has avg traffic delay.**
- **Delivery performance is influenced by both regional factors and route-level traffic conditions.**
- **KPI monitoring should be done to improve operational efficiency.**

Task 8

Task 8 output and insights:-

- **Ppt slide is prepared with queries input and screenshots, output table of each tasks following with the visualization of the tasks done.**
- **Key findings and Insights are provided for digestable understanding.**
- **Along with the pdf file.**
- **Videolink is being shared in the slide for the entire project work explanation.**

VIDEOLINK:-

https://drive.google.com/drive/folders/1jPSwjCmhDH0mSka6uRBAC1HI6sIsub5d?usp=drive_link

Thank You